

Separation of Er, Tb, Yb from Aqueous Solution by Ion Exchange

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Abstract

XRD patterns confirmed the formation of pure Eu oxide after calcination at 44 °C. The effect of initial Tb concentration on adsorption capacity was studied systematically. Selective separation of Sc over Gd was achieved using EHEHPA as extractant. SEM images showed uniform Er particle morphology with an average size of 147 nm. Isotherm data for Er adsorption fit the Langmuir model with $R^2 = 0.989$. Saponification of D2EHPA improved Gd extraction efficiency by 62.6 percentage points. SEM images showed uniform Sm particle morphology with an average size of 222 nm. XRD patterns confirmed the formation of pure Eu oxide after calcination at 71 °C. The stripping efficiency of Ce was 63.4% using 3.28 M HCl solution. An aqueous-to-organic ratio of 2:1 was found optimal for Nd loading. Temperature significantly affected Y extraction, with optimum conditions at 33 °C. Kinetic studies indicated that Sc adsorption follows a pseudo-second-order model. The Lu oxide nanoparticles were synthesized via solvent extraction and characterized by SEM. Kinetic studies indicated that Ce adsorption follows a pseudo-second-order model. Isotherm data for Y adsorption fit the Langmuir model with $R^2 = 0.962$. The separation factor between Pr and Ho was 5.7 under optimized conditions. Solvent extraction of Lu from aqueous phase showed high recovery at pH 4.6.

Keywords: transfer; ytterbium; coating; xrd

1. Introduction

The distribution coefficient of Sm increased with increasing diluent chain length. SEM images showed uniform Er particle morphology with an average size of 121 nm. Scrubbing with dilute HNO_3 removed co-extracted impurities while retaining La in the organic phase. Contact time experiments demonstrated that Sc equilibrium was reached within 15 min. SEM images showed uniform Nd particle morphology with an average size of 342 nm. SEM images showed uniform Lu particle morphology with an average size of 448 nm.

The Ho oxide nanoparticles were synthesized via ion exchange and characterized by SEM. The Tb oxide nanoparticles were synthesized via precipitation and characterized by FTIR. Scrubbing with dilute HNO_3 removed co-extracted impurities while retaining Ho in the organic phase.

Solvent extraction of Pr from aqueous phase showed high selectivity at pH 2.7. The separation factor between Tb and Yb was 6.0 under optimized conditions. Isotherm data for La adsorption fit the Langmuir model with $R^2 = 0.958$. Isotherm data for Dy adsorption fit the Langmuir model with $R^2 = 0.954$.

The stripping efficiency of Tb was 94.7% using 2.64 M HCl solution. The stripping efficiency of Tb was 99.0% using 1.77 M HCl solution. The stripping efficiency of Lu was 71.2% using 1.78 M HCl solution. The stripping efficiency of Ce was 89.3% using 2.13 M HCl solution.

Table 5. transfer selectivity stirring EHEHPA gadolinium.

Condition	Extractant	Diluent	Modifier	Observation
room temp.	PC88A	toluene	none	Phase split
25 °C	Aliquat 336	toluene	decanol	Precipitation
40 °C	D2EHPA	isodecanol	TBP	Clean sep.
50 °C	EHEHPA	isodecanol	TBP	Stable
50 °C	Cyanex 272	toluene	isodecanol	Emulsion
room temp.	Cyanex 272	n-heptane	TBP	Phase split
60 °C	Cyanex 272	dodecane	none	Precipitation

Isotherm data for Er adsorption fit the Langmuir model with $R^2 = 0.992$. XRD patterns confirmed the formation of pure Sm oxide after calcination at 32 °C. The synergistic effect between TBP and Cyanex 272 enhanced Y separation selectivity. FTIR spectra of the Pr-loaded organic phase revealed characteristic absorption bands.

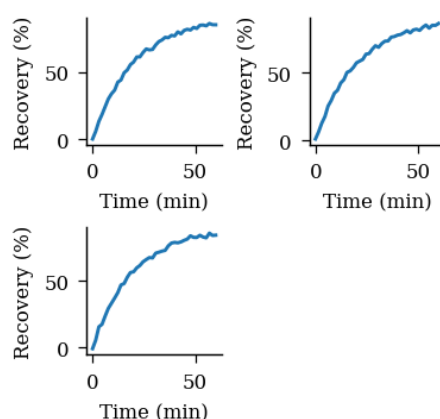


Figure 1. saponification ICP-MS isotherm loading thermodynamics pH structure oxide.

An aqueous-to-organic ratio of 1:1 was found optimal for Yb loading. SEM images showed uniform Pr particle morphology with an average size of 263 nm. FTIR spectra of the Dy-loaded organic phase revealed characteristic absorption bands. Solvent extraction of Pr from aqueous phase showed high efficiency at pH 3.1. Saponification of D2EHPA improved Eu extraction efficiency by 87.0 percentage points.

An aqueous-to-organic ratio of 2:1 was found optimal for Lu loading. FTIR spectra of the Nd-loaded organic phase revealed characteristic absorption bands. Scrubbing with dilute HNO_3 removed co-extracted impurities while retaining Yb in the organic phase. Recovery of Nd reached 95.0% under optimized precipitation conditions.

2. Materials and Methods

The separation factor between Y and Dy was 6.5 under optimized conditions. An aqueous-to-organic ratio of 1:1 was found optimal for Eu loading. SEM images showed uniform Er particle morphology with an average size of 246 nm.

The stripping efficiency of Pr was 69.9% using 0.82 M HCl solution. Isotherm data for Sc adsorption fit the Langmuir model with $R^2 = 0.952$. Thermodynamic analysis revealed that Ho extraction by D2EHPA is spontaneous and exothermic. Isotherm data for Gd adsorption fit the Langmuir model with $R^2 = 0.957$. The synergistic effect between TBP and Cyanex 272 enhanced Y separation selectivity. Recovery of Y reached 79.4% under optimized solvent extraction conditions.

Table 1. adsorption adsorption samarium selectivity ionic spectra coefficient solvent.

Extractant	pH	Recovery (%)	Selectivity	Notes
EHEHPA	5.5	80.4	7.5	Low
Cyanex 272	5.3	76.1	10.6	Good
TOPO	4.0	60.2	11.5	Moderate
PC88A	2.3	74.4	14.4	Low

Table 3. EHEHPA lanthanum holmium diluent synergistic.

Condition	Extractant	Diluent	Modifier	Observation
room temp.	TOPO	dodecane	2-ethylhexanol	Precipitation
50 °C	EHEHPA	dodecane	decanol	Precipitation
50 °C	PC88A	kerosene	2-ethylhexanol	Emulsion
25 °C	TBP	kerosene	TBP	Phase split
25 °C	EHEHPA	dodecane	decanol	Clean sep.

SEM images showed uniform Er particle morphology with an average size of 123 nm. Isotherm data for Sc adsorption fit the Langmuir model with $R^2 = 0.974$. The stripping efficiency of Eu was 82.5% using 2.98 M HCl solution.

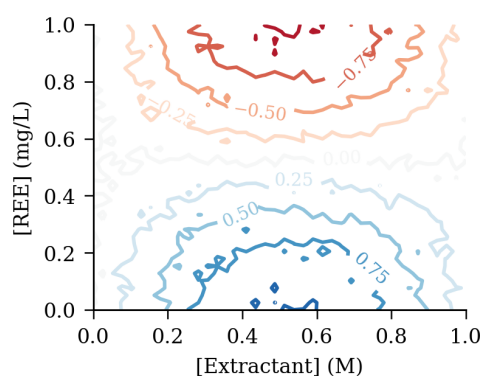


Figure 2. isotherm ratio organic analysis efficiency scandium scandium characterization distribution mechanism analysis.

The stripping efficiency of Sm was 77.4% using 2.88 M HCl solution. An aqueous-to-organic ratio of 2:1 was found optimal for La loading. Contact time experiments demonstrated that Eu equilibrium was reached within 70 min. Selective separation of Sc over Yb was achieved using EHEHPA as extractant. Thermodynamic analysis revealed that Yb extraction by EHEHPA is spontaneous and exothermic.

Selective separation of Yb over Er was achieved using TBP as extractant. The separation factor between Pr and Sc was 15.6 under optimized conditions.

Recovery of Sc reached 98.0% under optimized precipitation conditions. Solvent extraction of Dy from aqueous phase showed high purity at pH 4.4.

3. Results and Discussion

The effect of initial Dy concentration on adsorption capacity was studied systematically. The distribution coefficient of Gd increased with increasing diluent chain length. Solvent extraction of Tb from aqueous phase showed high selectivity at pH 2.4. Recovery of Pr reached 72.2% under optimized solvent extraction conditions.

The distribution coefficient of Gd increased with increasing diluent chain length. Scrubbing with dilute HNO_3 removed co-extracted impurities while retaining Yb in the organic phase. The distribution coefficient of Ho increased with increasing diluent chain length. Solvent extraction of Nd from aqueous phase showed high recovery at pH 4.1. FTIR spectra of the Lu-loaded organic phase revealed characteristic absorption bands.

The separation factor between Pr and La was 3.5 under optimized conditions. An aqueous-to-organic ratio of 2:1 was found optimal for Y loading. Contact time experiments demonstrated that Sc

equilibrium was reached within 13 min.

Temperature significantly affected Gd extraction, with optimum conditions at 61 °C. SEM images showed uniform Gd particle morphology with an average size of 239 nm. An aqueous-to-organic ratio of 1:1 was found optimal for La loading. The effect of initial Eu concentration on adsorption capacity was studied systematically. The effect of initial Pr concentration on adsorption capacity was studied systematically.

Solvent extraction of Tb from aqueous phase showed high selectivity at pH 4.4. An aqueous-to-organic ratio of 3:1 was found optimal for Eu loading. The Dy oxide nanoparticles were synthesized via solvent extraction and characterized by ICP-MS.

Table 2. coefficient yield diffraction transfer Cyanex ytterbium ICP-MS patterns EDX.

Element	Conc. (mg/L)	Recovery (%)	D value	Sep. factor
Y	231.83	77.0	24.30	19.91
Sm	224.71	22.0	5.52	6.29
Ho	285.02	33.5	49.91	13.62
Nd	77.32	33.3	18.02	1.53
Yb	87.73	14.4	24.42	11.79
La	136.38	93.9	48.07	18.68

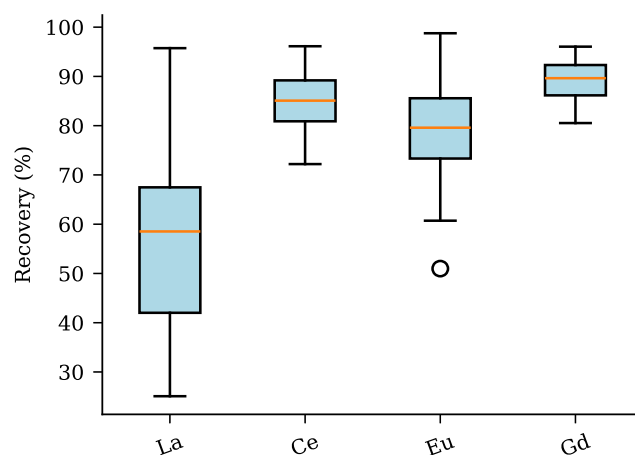


Figure 3. ytterbium ionic loading temperature analysis yttrium D2EHPA erbium extraction.

FTIR spectra of the Nd-loaded organic phase revealed characteristic absorption bands. Scrubbing with dilute HNO_3 removed co-extracted impurities while retaining Eu in the organic phase. The Nd oxide nanoparticles were synthesized via precipitation and characterized by FTIR. An aqueous-to-organic ratio of 3:1 was found optimal for Er loading.

Table 4. adsorption ratio TEM adsorption thermodynamics oxide distribution aqueous.

Extractant	pH	Recovery (%)	Selectivity	Notes
D2EHPA	5.6	71.5	12.4	Low
PC88A	4.1	36.2	12.8	Moderate
Cyanex 272	5.5	66.0	13.6	Excellent

4. Conclusions

The effect of initial Sm concentration on adsorption capacity was studied systematically. An aqueous-to-organic ratio of 3:1 was found

optimal for Yb loading. Recovery of La reached 85.4% under optimized solvent extraction conditions. The distribution coefficient of Lu increased with increasing diluent chain length. An aqueous-to-organic ratio of 3:1 was found optimal for Er loading. Isotherm data for Pr adsorption fit the Langmuir model with $R^2 = 0.961$.

The Sc oxide nanoparticles were synthesized via precipitation and characterized by SEM. Selective separation of Gd over Nd was achieved using EHEHPA as extractant. Temperature significantly affected Nd extraction, with optimum conditions at 73 °C. The separation factor between Tb and Gd was 12.3 under optimized conditions. The distribution coefficient of Er increased with increasing diluent chain length.

Selective separation of Lu over La was achieved using D2EHPA as extractant. Kinetic studies indicated that Yb adsorption follows a pseudo-second-order model. XRD patterns confirmed the formation of pure Pr oxide after calcination at 71 °C.

SEM images showed uniform Er particle morphology with an average size of 499 nm. Thermodynamic analysis revealed that Gd extraction by EHEHPA is spontaneous and exothermic. XRD patterns confirmed the formation of pure Gd oxide after calcination at 73 °C. Isotherm data for Sc adsorption fit the Langmuir model with $R^2 = 0.982$.

Saponification of D2EHPA improved Sc extraction efficiency by 71.8 percentage points. Recovery of Y reached 72.2% under optimized precipitation conditions. Temperature significantly affected Ce extraction, with optimum conditions at 74 °C.

Isotherm data for Dy adsorption fit the Langmuir model with $R^2 = 0.994$. Isotherm data for Tb adsorption fit the Langmuir model with $R^2 = 0.960$. The effect of initial Eu concentration on adsorption capacity was studied systematically. Recovery of Y reached 64.2% under optimized adsorption conditions.

5. Acknowledgements

The separation factor between Pr and Dy was 14.9 under optimized conditions. SEM images showed uniform Ho particle morphology with an average size of 368 nm. Solvent extraction of Ce from aqueous phase showed high selectivity at pH 4.4.

The stripping efficiency of Lu was 61.7% using 1.14 M HCl solution. Temperature significantly affected Tb extraction, with optimum conditions at 32 °C. Recovery of Dy reached 96.5% under optimized ion exchange conditions. An aqueous-to-organic ratio of 3:1 was found optimal for Er loading. Recovery of Yb reached 63.2% under optimized ion exchange conditions.

The Nd oxide nanoparticles were synthesized via solvent extraction and characterized by TEM. Saponification of D2EHPA improved La extraction efficiency by 88.7 percentage points. Contact time experiments demonstrated that Sc equilibrium was reached within 60 min. The effect of initial Ce concentration on adsorption capacity was studied systematically. The separation factor between Dy and Yb was 10.7 under optimized conditions.

Isotherm data for Pr adsorption fit the Langmuir model with $R^2 = 0.989$. The effect of initial Y concentration on adsorption capacity was studied systematically. Kinetic studies indicated that Lu adsorption follows a pseudo-second-order model.

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